

Sydney Girls High School

2015

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 3 – 5

10 Marks

- Attempt Questions 1 – 10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II Pages 6 – 9

60 Marks

- Attempt Questions 11 – 14
- Answer on the blank paper provided
- Begin a new page for each question
- Allow about 1 hour and 45 minutes for this section

Name:

Teacher:

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format or the content of the 2015 HSC Examination Paper in this subject.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

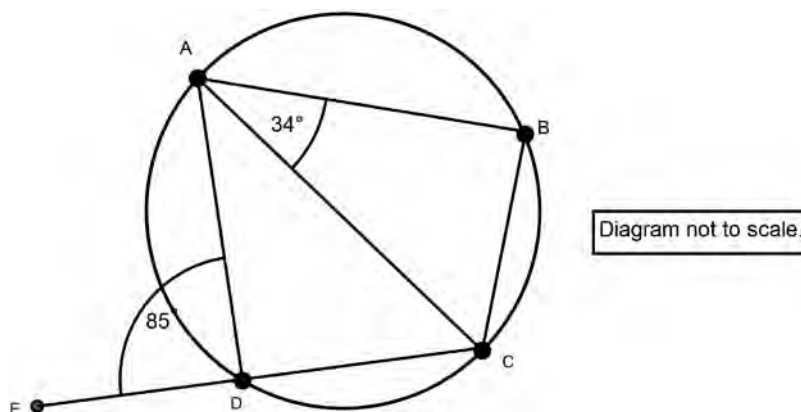
Use the multiple-choice answer sheet for Questions 1-10.

1. A committee of six is to be formed from seven women and nine men. Find the number of committees possible if exactly two members of the committee are to be men.

- (A) 1260
- (B) 2646
- (C) 36036
- (D) 60480

2. In the diagram below $\angle BAC = 34^\circ$ and $\angle ADE = 85^\circ$. What is the size of angle $\angle ACB$?

- (A) 51°
- (B) 56°
- (C) 61°
- (D) 60°



3. Use the substitution $u = e^x$ to determine which of the following is an expression for $\int \frac{e^x}{1+e^{2x}} dx$.

- (A) $\tan^{-1}(e^x) + C$
- (B) $\tan^{-1}(e^{2x}) + C$
- (C) $\frac{-1}{2(1+e^x)^2} + C$
- (D) $\frac{-e^x}{(1+e^x)^2} + C$

4. The radius of a spherical balloon is increasing at the rate of 2 cm/s. The rate at which the volume of the balloon is increasing when the radius is 10 cm is :
- (A) $200\pi \text{ cm}^3/\text{s}$
- (B) $400\pi \text{ cm}^3/\text{s}$
- (C) $800\pi \text{ cm}^3/\text{s}$
- (D) $100\pi \text{ cm}^3/\text{s}$
5. A stone is thrown vertically upwards with a speed of 21 m/s. How long is the stone in the air before it reaches its maximum height? (Assume acceleration due to gravity is 10 m/s^2 .)
- (A) 4.2 s
- (B) 0.48 s
- (C) 0.95 s
- (D) 2.1 s
6. The polynomial equation $f(x) = x^3 + x - 1$ has a root near $x = 0.5$. Using this as the initial approximation, determine another approximation (correct to four decimal places) to the root using one application of Newton's method.
- (A) $x = 0.7141$
- (B) $x = 0.7142$
- (C) $x = 0.7143$
- (D) $x = 0.7144$
7. If $y = \sin^{-1}\left(\frac{a}{x}\right)$, then $\frac{dy}{dx} =$
- (A) $\frac{-a}{\sqrt{x^2 - a^2}}$
- (B) $\frac{a}{\sqrt{x^2 - a^2}}$
- (C) $\frac{-a}{x\sqrt{x^2 - a^2}}$
- (D) $\frac{a}{x\sqrt{x^2 - a^2}}$

8. What is the value of $\sum_{k=1}^{20} {}^{20}C_k$?

- (A) 1 048 574
- (B) 1 048 575
- (C) 1 048 576
- (D) 1 048 577

9. Given that a , b and c are the roots of the equation $x^3 - 3x^2 + mx + 24 = 0$, and that $-a$ and $-b$ are the roots of the equation $x^2 + nx - 6 = 0$, then the value of n is :

- (A) 1
- (B) -1
- (C) 7
- (D) -7

10. The sum $1^4 + 2^4 + 3^4 + 4^4 + \dots + n^4$ is given by the expression $\frac{6n^5 + an^4 + bn^3 - n}{30}$.

The value of $a - b$ is :

- (A) -25
- (B) 25
- (C) -5
- (D) 5

Section II

Total 60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer all questions, starting each question on a new page.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Marks

(a) Solve $\frac{2}{x-5} \leq 3$. 3

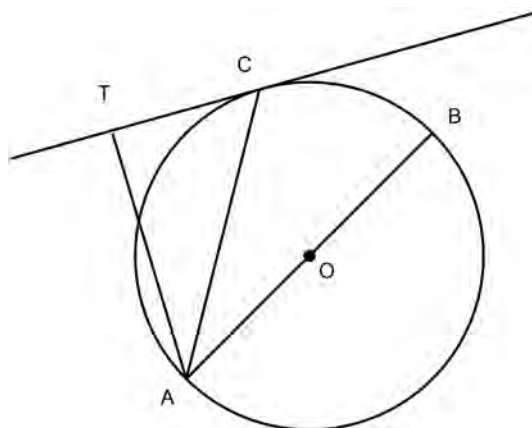
(b) Sketch the function $f(x) = \pi \cos^{-1}\left(\frac{x}{2}\right)$, clearly indicating the domain and range of the function. 3

(c) The velocity of a particle when x m from the origin is given by $v^2 = x^2 e^{3x} + 4$. 2
Find the acceleration of the particle when $x = 1$.

(d) Find the general solution to the equation $\sqrt{3} \tan \theta + 1 = 0$. 2

(e) Find the value of $\lim_{x \rightarrow 0} \frac{5x + \sin 3x}{2x}$. 2

- (f) In the diagram, AOB is the diameter of the circle with centre at O . TC is a tangent to the circle at the point C such that AC bisects $\angle TAB$. Copy the diagram onto your writing paper.
Prove that AT is perpendicular to TC .



End of Question 11

Question 12 (15 marks)**Marks**

- (a) By considering the derivative of $\ln(\tan x)$, find $\int \operatorname{cosec} 2x \, dx$. 3
- (b) In the expansion of $\left(x + \frac{2}{x^2}\right)^{10}$, find the coefficient of x . 3
- (c) $P(4p, 2p^2)$ and $Q(4q, 2q^2)$ are two variable points on the parabola $x^2 = 8y$. The tangents at P and Q intersect at the point T .
- (i) Derive the equation of the tangent at P . 1
- (ii) Hence, show that the point T has the coordinates $(2(p+q), 2pq)$. 2
- (iii) Given that $p^2 + q^2 = 10$, determine the cartesian equation of the locus of T . 2
- (d) Find $\int (\sin^2 x + 2\cos^2 x + 3\tan^2 x) \, dx$. 2
- (e) Prove that $\tan^{-1}(x+1) + \cot^{-1} x = \tan^{-1}(-x^2 - x - 1)$ for $x > 0$. 2

End of Question 12

Question 13 (15 marks)**Marks**

- (a) The growth rate per month of the number N of birds on a property during a drought is -20% of the excess of the bird population over 1000.
- (i) Express the information in the form of a differential equation and show that $N = 1000 + Ae^{-0.2t}$ (where t is the time in months) is a solution to this differential equation. 2
- (ii) Given that initially there are 8000 birds on the property, find the amount of time that will elapse before the population is reduced to half. 2
- (b) Ansett Airlines offer two options on all flights for their meal service – chicken or beef (vegetarians choose not to fly with Ansett). If 60% of the time Ansett passengers select the chicken dish, what is the probability that out of 7 randomly selected passengers at least 2 will select chicken for their meal? 2
- (c) An iPhone is thrown from the top of a building, 6 metres high, with an initial velocity of 8 m/s at an angle of 30° to the horizon.
- (i) Using 10 m/s^2 for acceleration due to gravity, derive the horizontal and vertical equations of motion for the iPhone. 2
- (ii) Determine the greatest height of the iPhone above ground level. 2
- (iii) Find the velocity and direction of the iPhone's path after 1 second. 2
- (d) Prove the following statement is true by mathematical induction for all integers $n \geq 1$. 3

$$\sum_{r=1}^n r(r!) = (n+1)! - 1$$

End of Question 13

Question 14 (15 marks)**Marks**

(a) A particle moves in a straight line with simple harmonic motion. At time t seconds, its displacement x metres from a fixed point O , is given by $x = 2 + 5 \sin\left(3t + \frac{\pi}{4}\right)$.

(i) Show that $\ddot{x} = -9(x - 2)$. 1

(ii) Determine the maximum speed of the particle and its displacement at this time. 2

(b) How many different arrangements of the word MAMMOTH can be made if only five letters are used? 2

(c) Use the substitution $u^2 = x + 1$ to find the volume of the solid formed by rotating the area bounded by the curve $y = \frac{x-1}{\sqrt{x+1}}$, the x axis and the lines $x = 3$ and $x = 8$ about the x axis. Express your answer in exact form. 4

(d) Use the expansion of $(1+x)^n$ to prove that 3

$$\frac{n+(-1)^n}{n+1} = \frac{1}{2} {}^n C_1 - \frac{1}{3} {}^n C_2 + \cdots + \frac{(-1)^n}{n} {}^n C_{n-1}.$$

(e) Given that $f(x) = Ax^3 + Bx^2 + Cx + D$ is a function with a double zero at $x = 1$, and with a minimum value of -4 when $x = -1$, find the values of A , B , C and D . 3

End of paper

2015 Trial MSC - Extension 1

SECTION I

1. ${}^9C_2 \times {}^7C_4 = 1260$ (A)

2. $\angle ABC = 85$ (ext. $\angle$ of cyclic quad)
 $\angle ACB + 34 + 85 = 180$ ($\angle$ sum of $\triangle ABC$)
 $\therefore \angle ACB = 61^\circ$ (C)

3. let $u = e^x$ $du = e^x dx$
 $\int \frac{e^x}{1+(e^x)^2} dx = \int \frac{du}{1+u^2} = \tan^{-1}u + C = \tan^{-1}(e^x) + C$ (A)

4. $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ $V = \frac{4}{3}\pi r^3$ $\frac{dV}{dr} = 4\pi r^2$
 $= 4\pi (10)^2 \times 2$
 $= 800\pi \text{ cm}^3/\text{s}$ (C)

5. when $t=0$, $y = 21 \text{ m/s}$ $t=?$ when $y=0$
 $\ddot{y} = -10$
 $\dot{y} = -10t + c$ $\therefore c = 21$
 $0 = -10t + 21$ $10t = 21$ $\therefore t = 2.1 \text{ s}$ (D)

6. $f(x) = x^3 + x - 1$ $f'(x) = 3x^2 + 1$
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{(0.5^3 + 0.5 - 1)}{3(0.5)^2 + 1}$
 $\doteq 0.7143$ (C)

7. $\frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{a^2}{x^2}}} \times \frac{-a}{x^2} = \frac{-a}{x\sqrt{x^2-a^2}}$ (C)

8. Consider $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$
 let $x=1$
 $(1+1)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$
 let $n=20$
 $\therefore \sum_{k=1}^{20} {}^{20}C_k + {}^{20}C_0 = (2)^{20}$
 $\therefore \sum_{k=1}^{20} {}^{20}C_k = 2^{20} - 1 = 1048575$ (B)

9. $a+b+c = 3$ $abc = -24$
 $-a-b = -n$ $ab = -6$
 $a+b = n$ $\therefore c = 4$
 $\therefore n = 3 - c = 3 - 4$
 $\therefore n = -1$ (B)

10. $1^4 = 6 + a + b - 1$ $a+b = 30 - 5 = 25$ (1)
 $1^4 + 2^4 = \frac{6(2)^5 + a(2)^4 + b(2)^3 - 2}{30}$
 $190 + 16a + 8b = 510$
 $2a+b = \frac{320}{8} = 40$ (2)
 $a = 15, b = 10$
 $a-b = 5$ (D)

Question 11. (15 Marks)

a) $\frac{2}{x-5} \leq 3 \quad (x \neq 5)$

$$\frac{2(x-5)^2}{(x-5)} \leq 3(x-5)^2$$

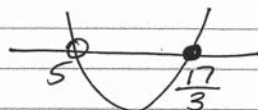
$$2(x-5) \leq 3(x-5)^2$$

$$3(x-5)^2 - 2(x-5) \geq 0$$

$$(x-5)[3(x-5)-2] \geq 0$$

$$(x-5)(3x-15-2) \geq 0$$

$$(x-5)(3x-17) \geq 0$$

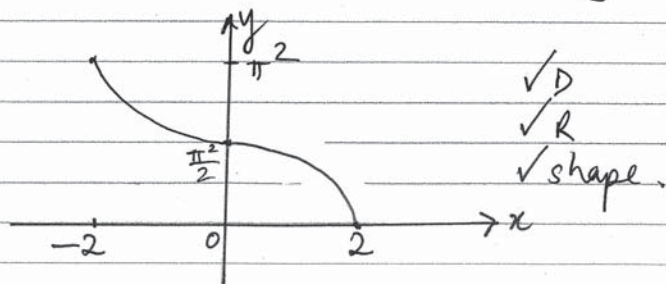


$$x \leq 5 \text{ or } x \geq \frac{17}{3} \quad (x \neq 5) \quad \checkmark \checkmark \checkmark$$

Many students lost a mark as $x \neq 5$.

b) $f(x) = \pi \cos^{-1}\left(\frac{x}{2}\right)$

Domain: $-1 \leq \frac{x}{2} \leq 1$ Range: $0 \leq \frac{y}{\pi} \leq \pi$
 $-2 \leq x \leq 2$ $0 \leq y \leq \pi$



c) $v^2 = x^2 e^{3x} + 4$

$$a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \frac{1}{2} (x^2 e^{3x} + 4)$$

$$= \frac{1}{2} [x^2 \cdot 3e^{3x} + e^{3x} \cdot 2x]$$

$$= \frac{1}{2} [3x^2 e^{3x} + 2x e^{3x}]$$

$$= \frac{1}{2} e^{3x} (3x^2 + 2x) \quad \checkmark$$

when $x=1$

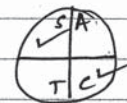
$$a = \frac{1}{2} \cdot e^3 \cdot (5)$$

$$a = \frac{5}{2} e^3 \text{ m/s}^2 \quad \checkmark$$

$$[a \approx 50.21 \text{ m/s}^2 \text{ (2 dec. pl.)}]$$

d) $\sqrt{3} \tan \theta = -1$

$$\tan \theta = -\frac{1}{\sqrt{3}}$$



$$\theta = \pi n + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

$$\theta = n\pi - \frac{\pi}{6} \quad (n \text{ is an integer})$$

OR $\theta = n\pi + \frac{5\pi}{6}$

$$e) \lim_{x \rightarrow 0} \frac{5x + \sin 3x}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{5x}{2x} + \lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$$

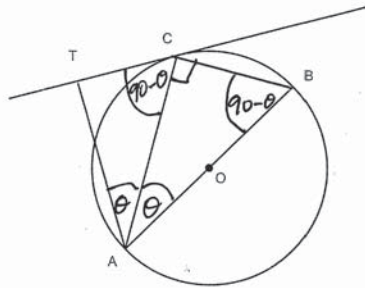
$$= \frac{5}{2} + \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \quad (2)$$

$$= \frac{8}{2}$$

$$= 4$$

* Some confusion for students with this limit.

f)



* Many more ways to prove that $AT \perp TC$

* The proof had to be logical from start to finish.

Join B to C

$\angle ACB = 90^\circ$ ($\angle$ in a semi circle)

Let $\angle TAC = \angle CAB = \theta$ (AC bisects $\angle TAB$)

In $\triangle ACB$: $\angle ABC = 90 - \theta$ ($\angle$ sum of $\triangle ACB$)

$\angle TCA = \angle ABC = 90 - \theta$ ($\angle$ in alternate segment)

In $\triangle ATC$: $\angle ATC + \angle TAC + \angle TCA = 180^\circ$ ($\angle$ sum $\triangle ATC$)

$$\angle ATC + \theta + 90 - \theta = 180$$

$$\angle ATC + 90 = 180$$

$$\angle ATC = 90^\circ$$

$\therefore AT \perp TC$

(3)

Q12 a) $y = \ln(\tan x)$
 $y' = \frac{\sec^2 x}{\tan x} = \frac{\frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x}} = \frac{1}{\cos x \sin x}$ ✓

Now $\frac{1}{\cos x \sin x} = \frac{2}{2 \sin x \cos x} = \frac{2}{\sin 2x} = 2 \operatorname{cosec} 2x$ ✓

$\frac{1}{2} \int 2 \operatorname{cosec} 2x dx = \frac{\ln[\tan x]}{2} + C$
 [Some students could not find $\frac{d[\ln(\tan x)]}{dx}$, hence couldn't get the final Ans.]

b) $(x + \frac{2}{x^2})^{10}$

$T_{k+1} = {}^{10}C_k \cdot a^{10-k} \cdot b^k$
 $= {}^{10}C_k \cdot x^{10-k} \cdot (2x^{-2})^k$ ✓
 $= {}^{10}C_k \cdot x^{10-k} \cdot 2^k \cdot x^{-2k}$
 $= {}^{10}C_k \cdot 2^k \cdot x^{10-3k}$

Equating the coefficients of x

$\therefore 10 - 3k = 1$
 $3k = 9$ ✓
 $k = 3$

The coefficient of x is

${}^{10}C_3 \cdot 2^3 = 960$ ✓

[Most students did well in this question.]

Q12 c) $P(4p, 2p^2) \quad Q(4q, 2q^2)$

i) $x^2 = 8y$
 $y = \frac{x^2}{8} \therefore y' = \frac{x}{4}$

At $P(4p, 2p^2) \therefore y' = \frac{4p}{4} = p$

Equation of the tangent at P.

$y - 2p^2 = p(x - 4p)$

$y = px - 2p^2$ ✓ ①

Similarly the equation of the tangent at Q

$y = qx - 2q^2$ ✓ ②

ii) solve ① and ②

$px - 2p^2 = qx - 2q^2$

$(p - q)x = 2(p + q)(p - q)$ ✓

$x = 2(p + q)$ Sub into ①

$y = p[2(p + q)] - 2p^2$

$y = 2p^2 + 2pq - 2p^2$

$y = 2pq$

Thus $T(2(p + q), 2pq)$ ✓

[Almost every one did well in parts (c/i) and (c/ii)]

Q12

c/iii) $x = 2(p+q)$ Given $p^2 + q^2 = 10$
 $y = 2pq$

$$x^2 = [2(p+q)]^2 \checkmark$$

$$x^2 = 4(p^2 + q^2 + 2pq)$$

$$x^2 = 4(10 + y)$$

$$x^2 = 4(y + 10) : \text{Locus of } T \checkmark$$

d) $\int (\sin^2 x + 2\cos^2 x + 3\tan^2 x) dx$

$$I = \int [\sin^2 x + \cos^2 x + \cos^2 x + 3(\sec^2 x - 1)] dx \checkmark$$

$$I = \int [3\sec^2 x + \frac{1}{2}(1 + \cos 2x) - 2] dx$$

$$I = \int (3\sec^2 x + \frac{1}{2}\cos 2x - \frac{3}{2}) dx$$

$$= 3\tan x + \frac{1}{4}\sin 2x - \frac{3}{2}x + C \checkmark$$

[A number of students forgot Trigonometric identities, double-angle formula, hence could not integrate this question correctly.]

Q12

e) Let $A = \tan^{-1}(x+1)$, Let $B = \cot^{-1}x$
 $\tan A = x+1$ $\cot B = x$
 $\tan B = \frac{1}{x}$

$$\tan^{-1}[-(x^2+x+1)] = -\tan^{-1}(x^2+x+1) \checkmark$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$= \frac{x+1 + \frac{1}{x}}{1 - (x+1) \times \frac{1}{x}}$$

$$= \frac{\frac{x^2+x+1}{x}}{\frac{x-x-1}{x}} = -(x^2+x+1)$$

Thus $A+B = \tan^{-1}[-(x^2+x+1)] \checkmark$

OR $\tan^{-1}(x+1) + \cot^{-1}x = \tan^{-1}(-x^2-x-1)$

[Many students could not simplify $\tan^{-1}(x+1)$, $\cot^{-1}x$ and $\tan[-(x^2+x+1)]$ correctly]

Q13 Ext 1 2015

a) i) $\frac{dN}{dt} = k(N - 1000)$

$= -0.2(N - 1000)$

$N = 1000 + A e^{-0.2t}$

$\frac{dN}{dt} = -0.2 A e^{-0.2t}$
 $A e^{-0.2t} = N - 1000$
 $= -0.2(N - 1000)$

* The setting out for this question was not very good. For a show question you need to show all the steps.

ii) at $t=0$, $N=8000$

$8000 = 1000 + A e^0$

$\therefore A = 7000$

$N = 1000 + 7000 e^{-0.2t}$

$4000 = 1000 + 7000 e^{-0.2t}$

$\frac{3}{7} = e^{-0.2t}$

$\ln\left(\frac{3}{7}\right) = -0.2t$

$t = 4.23$ months

* This question was done very well.

b) $P(\text{at least 2 chicken}) =$

$1 - {}^7C_1 (0.6)(0.4)^6 - {}^7C_0 (0.6)^0 (0.4)^7$

$= 1 - \frac{1344}{78125} - \frac{128}{78125}$

$= 0.98$

* Many students couldn't do this question.

The particle is travelling at 9.2 m/s down wards at $\theta = 41^\circ$ to the horizontal.

* many students didn't know how to find the direction

c)

i)

$\ddot{x} = 0$

$\dot{x} = C$

$x = V \cos \theta$

$x = Vt \cos \theta + C$

at $t=0$, $x=0$

$\therefore C=0$

$x = Vt \cos \theta$

$= 8t \cos 30^\circ$

$= 8t \times \frac{\sqrt{3}}{2}$

$x = 4\sqrt{3}t^2$

* Many students didn't show all the steps of deriving these equations

$\ddot{y} = -10$

$\dot{y} = -10t + C$

$C = V \sin \theta$

$\dot{y} = -10t + V \sin \theta$

$y = -5t^2 + Vt \sin \theta + C$

at $t=0$, $C=6$

$\therefore y=6$

$y = -5t^2 + Vt \sin \theta + 6$

$= -5t^2 + 8t \times \frac{1}{2} + 6$

$y = -5t^2 + 4t + 6$

ii) $\dot{y} = 0$

$-10t + 4 = 0$

$t = \frac{2}{5}$

* This question was done well

$y = -5\left(\frac{2}{5}\right)^2 + 8\left(\frac{2}{5}\right) + 6$

$y = 6.8$

iii) $\dot{x} = 4\sqrt{3}$

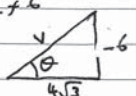
$\dot{y} = -10\left(\frac{2}{5}\right) + 8\left(\frac{1}{2}\right)$

$= -6$

$v^2 = (4\sqrt{3})^2 + 36$

$v = 9.2 \text{ m/s}$

$\tan \theta = \frac{6}{4\sqrt{3}} \therefore \theta = 41^\circ$



d)

$1 + 2(2!) + 3(3!) + \dots + n(n!) = (n+1)! - 1$

when $n=1$

LHS = 1

RHS = $2! - 1$

$= 1$

$\therefore$ true for $n=1$

when

$n=k$

$1 + 2(2!) + \dots + k(k!) = (k+1)! - 1$

prove true for $n=k+1$

$1 + 2(2!) + \dots + k(k!) + (k+1)(k+1)! = (k+2)! - 1$

LHS = $(k+1)! - 1 + (k+1)(k+1)!$

$= (k+1)! [1 + k+1] - 1$

$= (k+1)! (k+2) - 1$

$= (k+2)! - 1$

$= \text{RHS}$

$\therefore$ true for $n \geq 1$

By Mathematical induction

* This question was done well, except some students didn't show the last step properly $\therefore$ lost mark.

Question 14 (15 Marks)

a) $x = 2 + 5 \sin(3t + \frac{\pi}{4}) \Rightarrow x - 2 = 5 \sin(3t + \frac{\pi}{4})$

i) $\dot{x} = 0 + 5 \cos(3t + \frac{\pi}{4}) \times 3$
 $= 15 \cos(3t + \frac{\pi}{4})$

$$\ddot{x} = -15 \sin(3t + \frac{\pi}{4}) \times 3$$

$$= -45 \sin(3t + \frac{\pi}{4})$$

$$= -9 [5 \sin(3t + \frac{\pi}{4})]$$

$$\therefore \ddot{x} = -9(x - 2), \text{ as required.}$$

ii) Maximum speed occurs at the centre of motion i.e. at $x = 2$

When $x = 2$: $5 \sin(3t + \frac{\pi}{4}) = 0$
 $3t + \frac{\pi}{4} = \pi \quad (t > 0)$

$$3t = \frac{3\pi}{4}$$

$$t = \frac{\pi}{4} \text{ sec.}$$

At $x = 2$: $\dot{x} = 15 \cos(\frac{3\pi}{4} + \frac{\pi}{4})$

$$= 15 \cos \pi$$

$$= -15 \text{ m/sec.}$$

$$\therefore |\dot{x}|_{\max} = 15 \text{ m/sec at } x = 2.$$

b). different arrangements of the word MAMMOTH using only 5 letters.

Case 1: (1M) M _ _ _ _ = $5!$ ways.

Case 2: (2M's) M M _ _ _ = $\frac{5!}{2!} \times {}^4C_3$

Case 3: (3M's) M M M _ _ = $\frac{5!}{3!} \times {}^4C_2$

Total arrangements

$$= 5! + \frac{5!}{2!} \times {}^4C_3 + \frac{5!}{3!} \times {}^4C_2$$

$$= 120 + 60 \times 4 + 20 \times 6$$

$$= 120 + 240 + 120$$

$$= \underline{\underline{480.}}$$

★ One mark awarded for considering the different cases of repetition.
One mark for the correct answer.

$$c) V = \pi \int_3^8 y^2 dx$$

$$= \pi \int_3^8 \frac{(x-1)^2}{(x+1)} dx$$

$$\begin{aligned} u^2 &= x+1 \\ 2u du &= dx \end{aligned}$$

$$= \pi \int_2^3 \frac{(u^2-2)^2}{u^2} \cdot 2u du$$

$$\begin{cases} u^2-1 = x \\ u^2-2 = x-1 \end{cases}$$

$$= 2\pi \int_2^3 \left(\frac{u^4}{u} - \frac{4u^2}{u} + \frac{4}{u} \right) du \quad \text{when } x=8, u^2=9 \\ u=3$$

$$= 2\pi \int_2^3 \left(u^3 - 4u + \frac{4}{u} \right) du \quad x=3, u^2=4 \\ u=2$$

$$= 2\pi \left[\frac{u^4}{4} - 2u^2 + 4\ln u \right]_2^3$$

$$= 2\pi \left[\left(\frac{81}{4} - 18 + 4\ln 3 \right) - \left(4 - 8 + 4\ln 2 \right) \right]$$

$$= 2\pi \left[\frac{25}{4} + 4\ln 3 - 4\ln 2 \right]$$

$$= \frac{25\pi}{2} + 8\pi(\ln 3 - \ln 2) \text{ units}^3.$$

* Most students substituted incorrectly with the new variable, back into the integral.

d)

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$$

Integrate both sides w.r.t. x :

$$\frac{(1+x)^{n+1}}{n+1} + C = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots + \frac{{}^nC_{n-1} x^n}{n} + \frac{{}^nC_n x^{n+1}}{n+1}$$

Let $x=0$:

$$\frac{1}{n+1} + C = 0 \quad \therefore C = -\frac{1}{n+1}$$

$$\therefore \frac{(1+x)^{n+1} - 1}{(n+1)} = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots + \frac{{}^nC_{n-1} x^n}{n} + \frac{{}^nC_n x^{n+1}}{n+1}$$

Let $x=-1$: noting that ${}^nC_0 = {}^nC_n = 1$

$$-\frac{1}{n+1} = -1 + \frac{1}{2} {}^nC_1 - \frac{1}{3} {}^nC_2 + \dots + \frac{(-1)^n}{n} {}^nC_{n-1} + \frac{(-1)^{n+1}}{n+1}$$

$$1 - \frac{1}{n+1} - \frac{(-1)(-1)^n}{n+1} = \frac{1}{2} {}^nC_1 - \frac{1}{3} {}^nC_2 + \dots + \frac{(-1)^n}{n} {}^nC_{n-1}$$

$$\frac{n+1-1}{n+1} + \frac{(-1)^n}{n+1} = \frac{1}{2} {}^nC_1 - \frac{1}{3} {}^nC_2 + \dots + \frac{(-1)^n}{n} {}^nC_{n-1}$$

$$\therefore \frac{n+(-1)^n}{n+1} = \frac{1}{2} {}^nC_1 - \frac{1}{3} {}^nC_2 + \dots + \frac{(-1)^n}{n} {}^nC_{n-1}$$

* Most students forgot the constant of integration in the first step.

$$e) \quad f(x) = Ax^3 + Bx^2 + Cx + D$$

$$f'(x) = 3Ax^2 + 2Bx + C$$

$$f(1) = 0 : A + B + C + D = 0 \dots (1)$$

$$f'(1) = 0 : 3A + 2B + C = 0 \dots (2)$$

$$f'(-1) = 0 : 3A - 2B + C = 0 \dots (3)$$

$$f(-1) = -4 : -A + B - C + D = -4 \dots (4)$$

$$(1) + (4) : 2B + 2D = -4 \dots (5)$$

$$(2) - (3) : 4B = 0$$

$$\underline{\underline{B = 0}}$$

$$\text{From (5) : } 2D = -4$$

$$\underline{\underline{D = -2}}$$

$$\text{From (1) : } A + C - 2 = 0 \dots (6)$$

$$\text{From (2) : } 3A + C = 0 \dots (7)$$

$$(7) - (6) : 2A + 2 = 0$$

$$2A = -2$$

$$\underline{\underline{A = -1}}$$

$$\text{From (6) : } -1 + C - 2 = 0$$

$$\underline{\underline{C = 3}}$$

$$\therefore A = -1, B = 0, C = 3, D = -2.$$